# Additional materials for “Machine Learning Pipeline for Malignant Breast Cancer Cell Prediction”

## Plots

This file contains supplementary visualizations that support the main analysis presented in the report. The included plots provide additional insights into feature distributions, class separation, and correlation structures, offering a more comprehensive understanding of the dataset beyond the tables discussed in the main text.

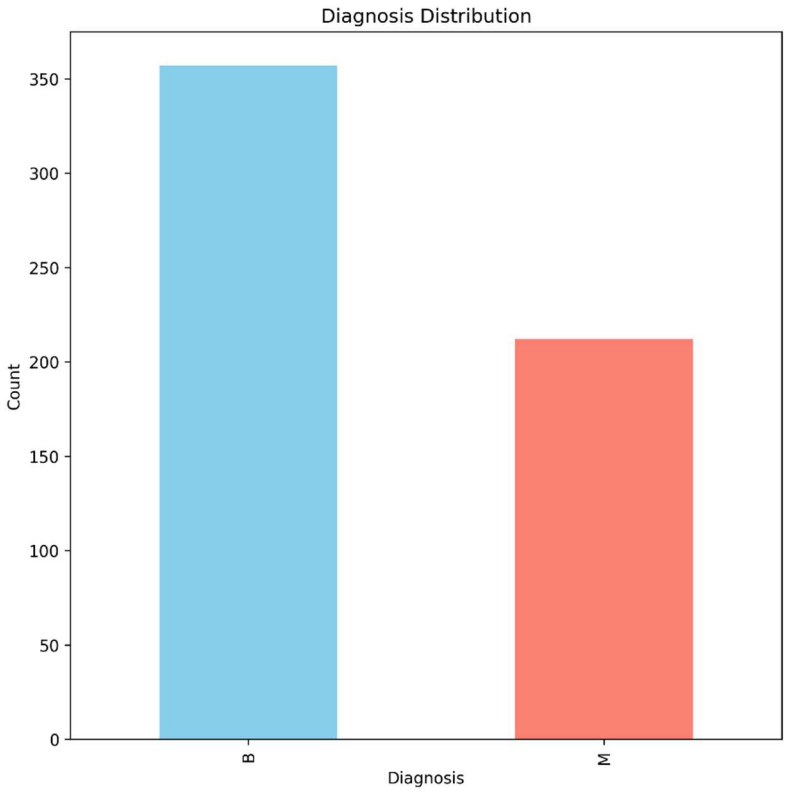


**Figure 1. Diagnosis distribution of the dataset:** Class distribution of the 569 samples, showing 357 benign (class 0) and 212 malignant (class 1) cases. Link: https://github.com/giacomo-timelli/AML_project_breast_cancer_classification/blob/main/eda/Diagnosis_distribution.png .

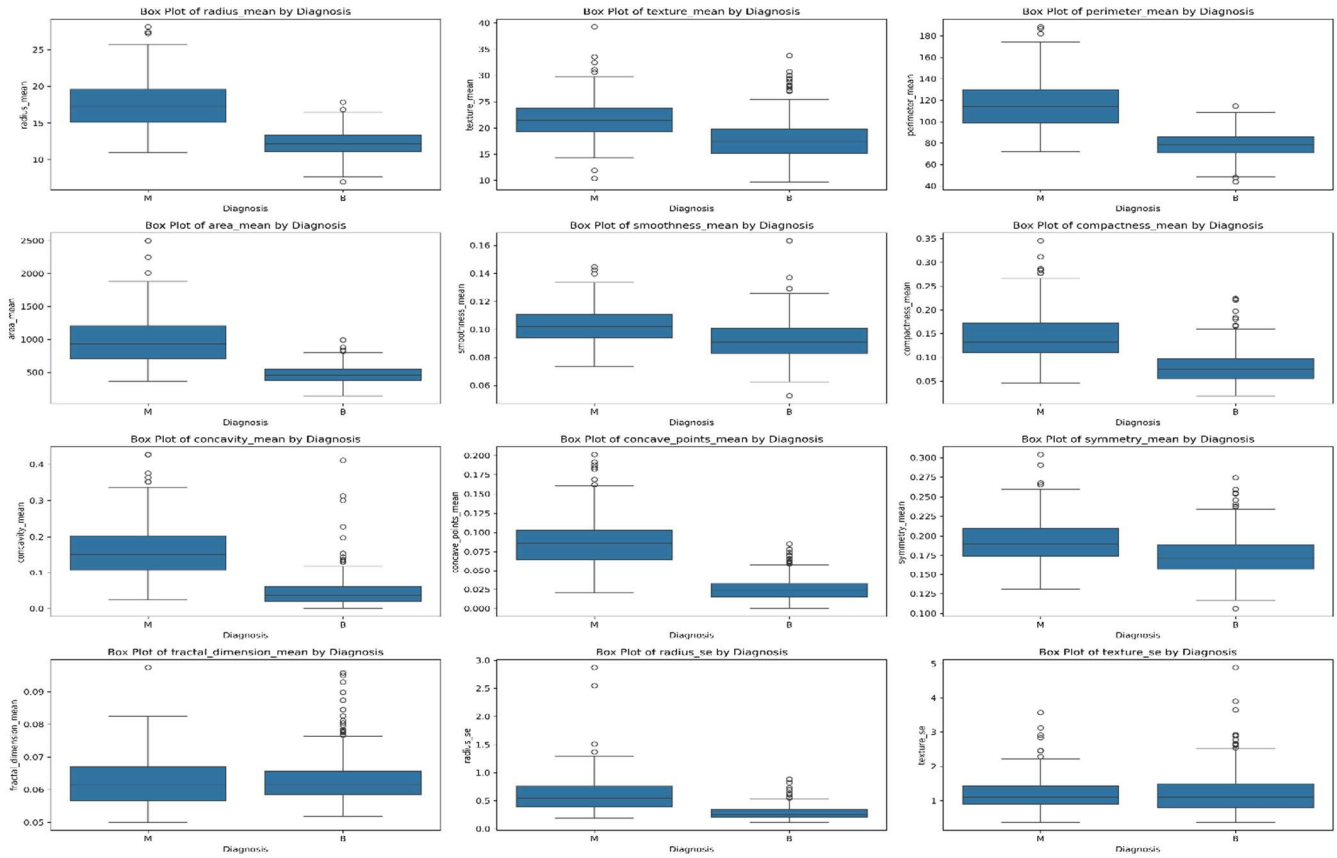


**Figure 2. Boxplot of feature distributions by diagnosis:** Distribution of each feature across benign and malignant classes, emphasizing variables with the highest discriminative power. Full plot at this link.

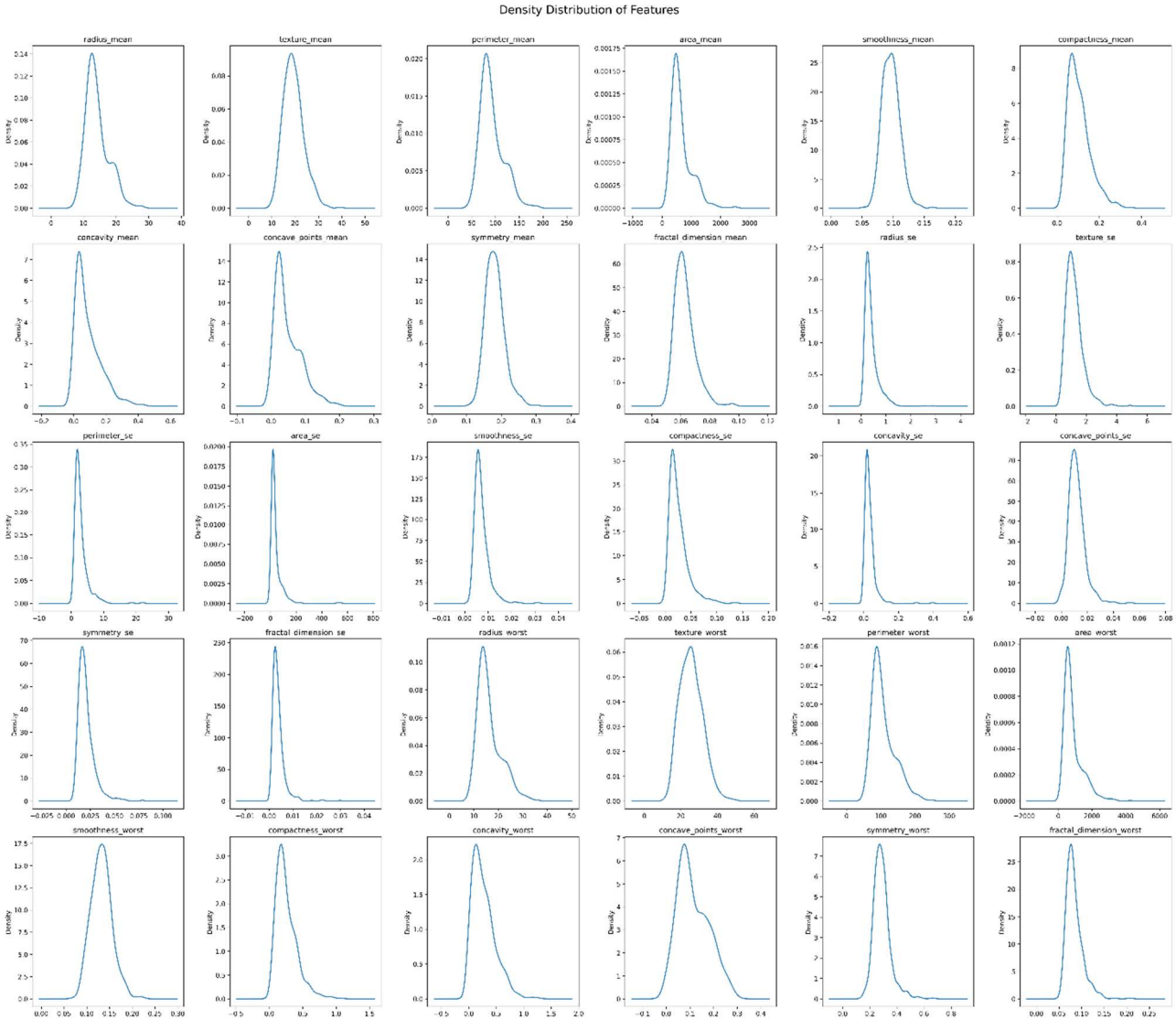


**Figure 3. Feature density distributions:** Density plots illustrating the distribution of selected features across samples (link to the plot).

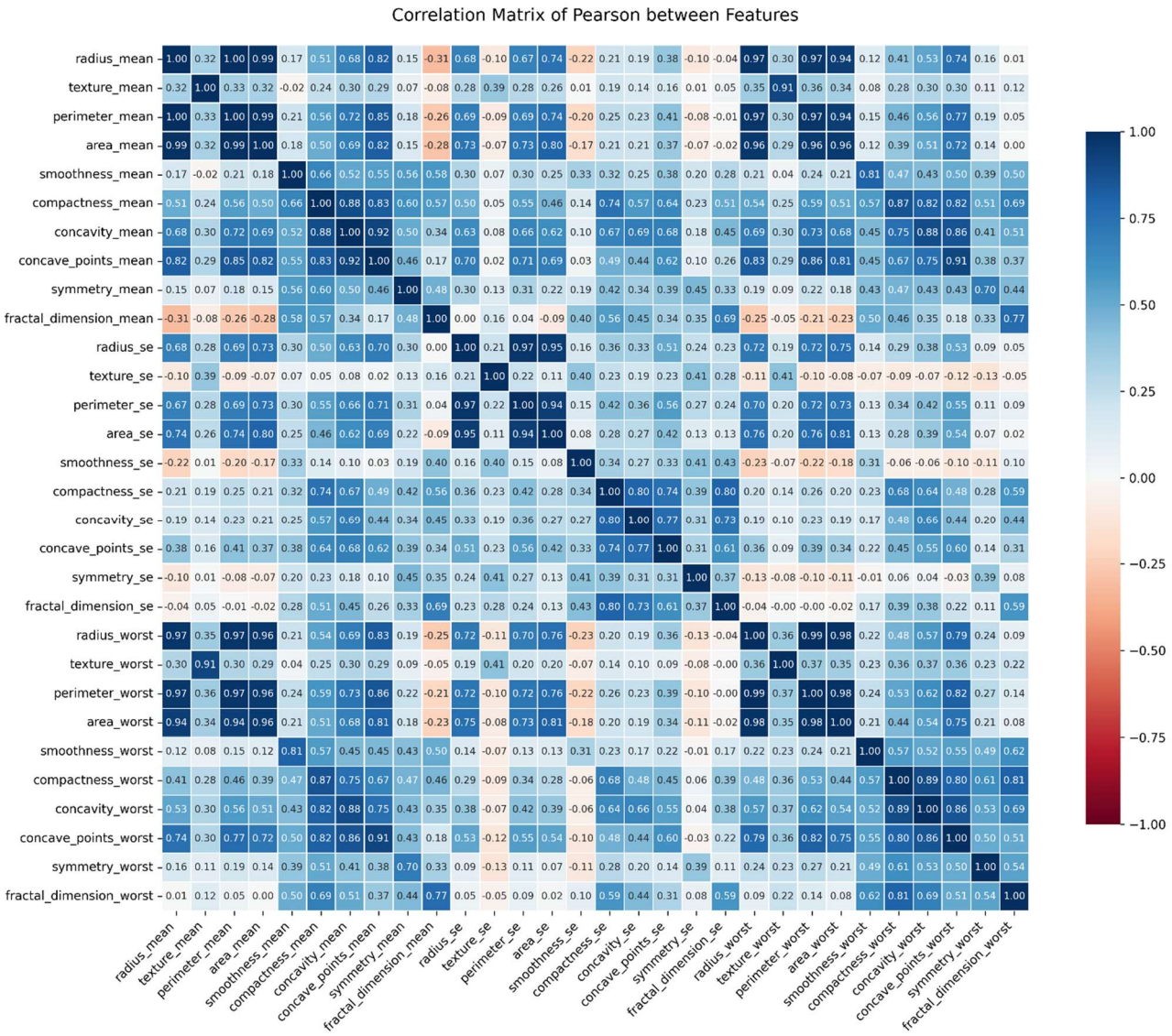


**Figure 4. Heatmap of Pearson correlations:** Pairwise linear correlations among the 30 features are visualized, with both axes representing the feature set. Correlation strength is encoded by a color scale, where darker blue indicates stronger positive correlations and red indicates negative correlations, as shown in the legend (link to the plot).

For the remaining plots and complete visual analysis, please refer to the GitHub repository.